

Mathematical Foundations and Academic References

Intelligent Multi-Sink WSN Optimization Framework — Technical Reference Guide

This document outlines the complete mathematical framework for the Intelligent Multi-Sink Wireless Sensor Network (WSN) Optimization Framework. It serves as a rigorous academic reference guide, bridging standard physical-layer baselines with novel reinforcement learning models and statistical validation methods.

Section 1: Foundational & Physics Models (Published Sources)

The following mathematical formulations are established baselines in the WSN literature. They are utilized verbatim to establish physical and statistical credibility.

1.1 The First-Order Radio Energy Model

- Role in Project: Used to track physical energy depletion (η) during transmission, reception, and data aggregation.

Energy to Transmit a k -bit message over distance d :

$$E_{Tx}(k, d) = k * E_{elec} + k * \epsilon_{fs} * d^2, \text{ if } d < d_0$$

$$E_{Tx}(k, d) = k * E_{elec} + k * \epsilon_{mp} * d^4, \text{ if } d \geq d_0$$

Energy to Receive a k -bit message:

$$E_{Rx}(k) = k * E_{elec}$$

Threshold Distance (d_0):

$$d_0 = \sqrt{\epsilon_{fs} / \epsilon_{mp}}$$

Data Aggregation Energy at Cluster Head:

$$E_{DA}(k) = k * E_{da_bit}$$

- Original Paper: Heinzelman, W. R., Chandrakasan, A., & Balakrishnan, H. (2000). Energy-efficient communication protocol for wireless microsensor networks. Proceedings of the 33rd Hawaii International Conference on System Sciences.
- Direct Access Link: <https://www.gta.ufrj.br/wsns/Routing/leach.pdf>

1.2 Topology and Centrality Baseline (HEED)

- Role in Project: Provides the baseline for using localized node neighborhood densities (V_i / N) to establish cluster head centrality, preventing spatial imbalances.
- Original Paper: Younis, O., & Fahmy, S. (2004). HEED: A hybrid, energy-efficient, distributed clustering approach for ad-hoc sensor networks. IEEE Transactions on Mobile Computing, 3(4), 366-379.

- Direct Access Link: <https://www.cs.purdue.edu/homes/fahmy/papers/tmc04.pdf>

1.3 Dynamic Environmental Threshold Model (EWMA & EWMV)

- Role in Project: Computes running means and variances of environmental data streams at the Edge/CH level to construct adaptive, noise-immune anomaly filtering windows.

Exponentially Weighted Moving Average (Mean μ_t):

$$\mu_t = (1 - \gamma) * \mu_{t-1} + \gamma * X_t$$

Exponentially Weighted Moving Variance (σ_t^2):

$$\sigma_t^2 = (1 - \gamma) * \sigma_{t-1}^2 + \gamma * (X_t - \mu_{t-1})^2$$

- Original Paper: Parhate, V. G., Sen, P. H., & Deshpande, L. (2026). AI-Driven Anomaly Detection in Wireless Sensor Networks. International Journal of Computer Information Systems and Industrial Management Applications, 18(3), 1292–1306.
- Direct Access Link: <https://cspub-ijcisim.org/index.php/ijcisim/article/view/2461>

1.4 Non-Parametric Performance Evaluation (Statistical Proof)

- Role in Project: Used to statistically validate that the proposed deep reinforcement learning (DQN) routing scheme outperforms baseline models (e.g., LEACH, PEGASIS) over dozens of simulation trials with high confidence.

Wilcoxon Signed-Rank Test (Comparing 1 baseline over N_r runs):

$$W = \sum_{i=1}^{N_r} [\text{sgn}(x_{1,i} - x_{2,i}) * R_i]$$

Friedman Test (Comparing multiple baselines simultaneously):

$$\chi_r^2 = [12 / (N * k * (k+1))] * \sum_{j=1}^k R_j^2 - 3N(k+1)$$

- Original Paper: Demšar, J. (2006). Statistical comparisons of classifiers over multiple data sets. Journal of Machine Learning Research, 7, 1-30.
- Direct Access Link: <https://www.jmlr.org/papers/volume7/demsar06a/demsar06a.pdf>

1.5 Foundations of Reinforcement Learning Packet Routing

- Role in Project: Establishes the framework for packet transmission optimization modeled as a Reinforcement Learning state-penalty system.
- Original Paper: Boyan, J. A., & Littman, M. L. (1994). Packet routing in dynamically changing networks: A reinforcement learning approach. Advances in Neural Information Processing Systems, 6, 671-678.
- Direct Access Link: https://publications.ri.cmu.edu/storage/publications/pub_files/pub1/boyan_justin_1994_1/boyan_justin_1994_1.pdf

Section 2: Custom & Adapted Models (Our Novel Contribution)

These equations represent the custom, multi-metric formulations designed specifically for the Intelligent Multi-Sink WSN architecture. In manuscripts, these should be framed as proposed or formulated models.

2.1 Cluster Head Selection Utility Metric (W_i)

Combines the normalized remaining battery energy with the local spatial neighborhood density ratio to elect high-energy, topologically central cluster heads.

$$W_i = \alpha * (E_{\text{residual}(i)} / E_{\text{max}}) + \beta * (V_i / N)$$

Subject to: $\alpha + \beta = 1$

2.2 Rate-of-Change Bypass Trigger (High Priority Queue)

Sensed data falling inside the normal fluctuation limit is discarded as redundant. An instant delta exceeding the rapid change limit bypasses the queue and is dispatched immediately.

$$\text{High Priority Bypass Trigger: } |X_t - X_{t-1}| > k_2 * \sigma_{t-1}$$

$$\text{Normal Data Filter Limit: } |X_t - \mu_{t-1}| < k_1 * \sigma_{t-1}$$

2.3 Multi-Metric Path Cost Function (C_j)

Evaluates the cost of path j by scaling physical distance, delay, congestion queue length, and link loss, while prioritizing high-energy local transmitter remaining budgets.

$$C_j = w_1 * D_j + w_2 * \tau_j + w_3 * Q_j + w_4 * (1 - L_j) - w_5 * E_{CH}$$

Subject to: $\sum_{n=1}^5 w_n = 1$

2.4 Custom DQN Reward Functions

Directs the dual-queue and multi-sink pathing agents to minimize latencies, packet drops, and transmission energy while maintaining buffer health.

Queue Management Reward (R_{queue}):

$R_{\text{queue}} = -P_{\text{drop_HP}}$, if a High Priority packet is dropped

$R_{\text{queue}} = -P_{\text{drop_LP}}$, if a Low Priority packet is dropped

$R_{\text{queue}} = \gamma_1 * E_{\text{residual}} - \gamma_2 * (Q_{\text{HP}} + \omega * Q_{\text{LP}})$, on successful transmission

Intelligent Routing Reward (R_{routing}):

$R = -[a * \text{Delay} + b * \text{Queue_Drop_Rate} + c * (1 - \text{Success_Rate})]$

Section 3: Essential Supporting Mathematical Gaps

To guarantee the mathematical framework is robust against strict academic review, the following baseline physical models are incorporated into the system model:

3.1 Buffer Queue Dynamics Model

$Q_{\text{HP}}(t+1) = \min(Q_{\text{max}}, Q_{\text{HP}}(t) + A_{\text{HP}}(t) - S_{\text{HP}}(t))$

$Q_{\text{LP}}(t+1) = \min(Q_{\text{max}}, Q_{\text{LP}}(t) + A_{\text{LP}}(t) - S_{\text{LP}}(t))$

- $Q(t)$: Current buffer queue length.
- $A(t)$: Incoming packet rate at step t .
- $S(t)$: Dispatched/served packets at step t .
- Q_{max} : Physical node memory limit.

3.2 SNR and Packet Delivery Ratio (L_j)

Connects physical-layer signal strength and bit error rates directly to the link quality metric utilized in the multi-metric routing cost evaluation.

$L_j = (1 - \text{BER})^k$

$\text{SNR} = 10 * \log_{10}(P_{\text{rx}} / N_0)$

3.3 Markov Decision Process Bellman Optimality Model

Formalizes the temporal-difference update paradigm of the DQN agents, showing how the neural network updates Q-values to maximize long-term sensor network operational lifetime.

$Q^*(s, a) = R(s, a) + \lambda * \max_{a'} Q^*(s', a')$

Section 4: BibTeX Reference Templates for LaTeX Manuscript

Copy and paste these templates directly into your bibliography file (.bib):

```
@inproceedings{heinzelman2000energy,  
  title={Energy-efficient communication protocol for wireless microsensor networks},  
  author={Heinzelman, Wendi Rabiner and Chandrakasan, Anantha and Balakrishnan, Hari},  
  booktitle={Proceedings of the 33rd Annual Hawaii International Conference on System Sciences},  
  pages={10--pp},  
  year={2000},  
  organization={IEEE}  
}
```

```
@article{younis2004heed,  
  title={HEED: a hybrid, energy-efficient, distributed clustering approach for ad-hoc sensor networks},  
  author={Younis, Ossama and Fahmy, Sonia},  
  journal={IEEE Transactions on Mobile Computing},  
  volume={3},  
  number={4},  
  pages={366--379},  
  year={2004},  
  publisher={IEEE}  
}
```

```
@article{parhate2026ai,  
  title={AI-Driven Anomaly Detection in Wireless Sensor Networks},  
  author={Parhate, Vivek G and Sen, Praveen H and Deshpande, Leena and others},  
  journal={International Journal of Computer Information Systems and Industrial Management Applications},  
  volume={18},  
  number={3},  
  pages={1292--1306},  
  year={2026}  
}
```

```
@inproceedings{boyan1994packet,  
  title={Packet routing in dynamically changing networks: A reinforcement learning approach},  
  author={Boyan, Justin A and Littman, Michael L},  
  booktitle={Advances in Neural Information Processing Systems},  
  pages={671--678},  
  year={1994}  
}
```

```
@article{demsar2006statistical,  
  title={Statistical comparisons of classifiers over multiple data sets},  
  author={Dem{\v{s}}ar, Janez},  
  journal={Journal of Machine Learning Research},  
  volume={7},  
  pages={1--30},  
  year={2006}  
}
```